

Leetcode

Functional requirements:

- user should be able to search for any problem based on his preference,
- user should be able to solve any problem based on his preference,
- user should be able to participate in any coding contest based on his preference

Non-functional requirements:

- data-partitioning,
- load balancing,
- fault tolerance,
- vertical scalability,
- horizontal scalability,
- low latency,
- high throughput ,
- CAP theorem (We need to prioritise Strong Consistency for rank predictions during contests & for search scenario we need to prioritise 99.99% uptime i,e Availability>>Consistency)

Core Entities:

- user,
- problem,
- contest,
- solution

API :

POST:/submit/solution

```
{
  userID,
  filemetadata
}
```

GET:/search/{problemID}

HLD :

